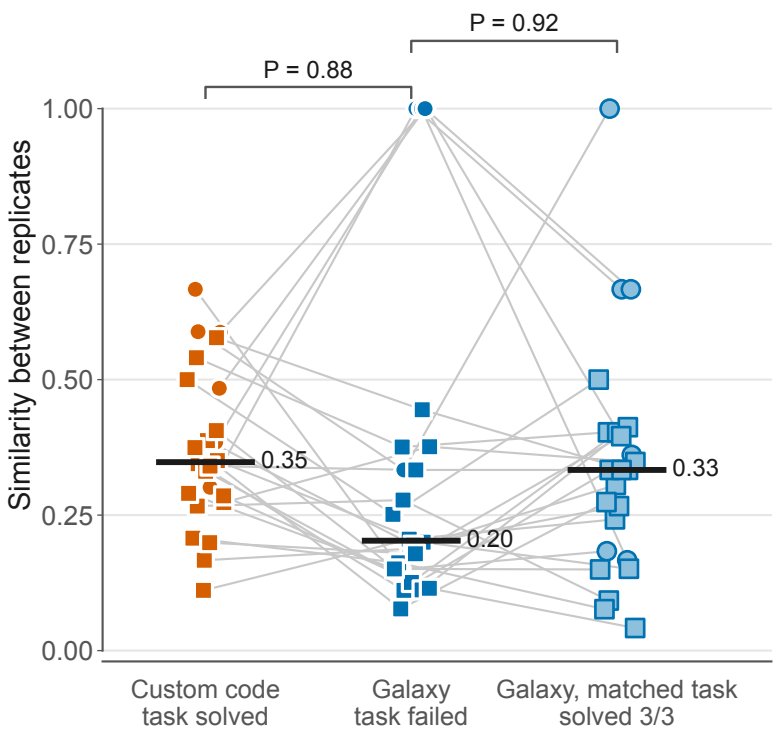
Do the three replicates of a task follow the same trajectory?

● BixBench-Verified-50 ■ CompBioBench

The 24 task × model pairs that only Galaxy fails (custom code solves ≥ 2 of 3 runs, Galaxy 0 or 1 of 3), compared with the custom-code runs of the same pairs and with Galaxy runs on matched tasks Galaxy solved 3/3 (same benchmark and model, closest median number of Galaxy steps). A step is one executed Galaxy job or command, labelled by tool or by script libraries. Each dot is one pair: the mean over its three replicate pairs. Lines join the same pair. P: paired cluster sign-flip test (clusters: tasks); black bars: medians.

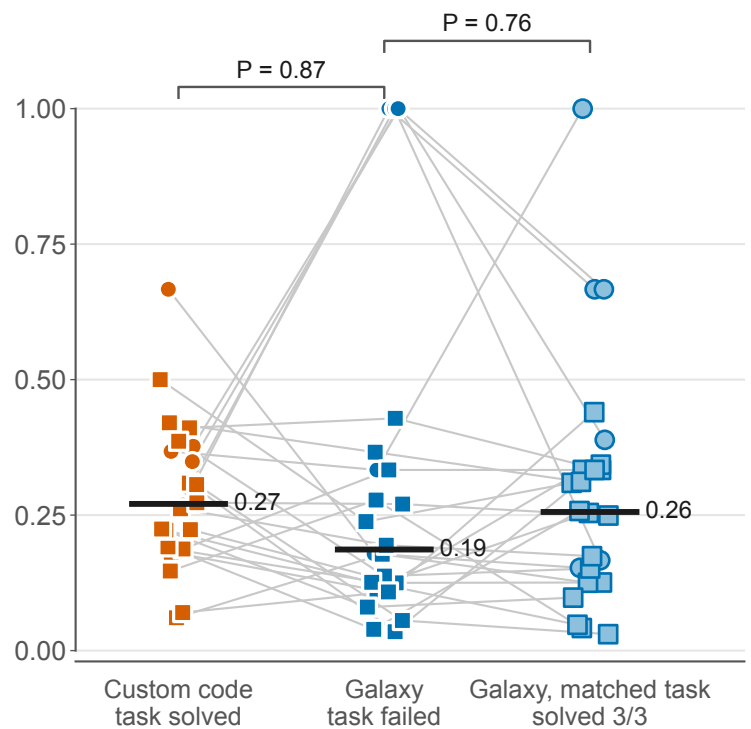
Approach similarity

Same tools and scripts
(Jaccard overlap of step labels)



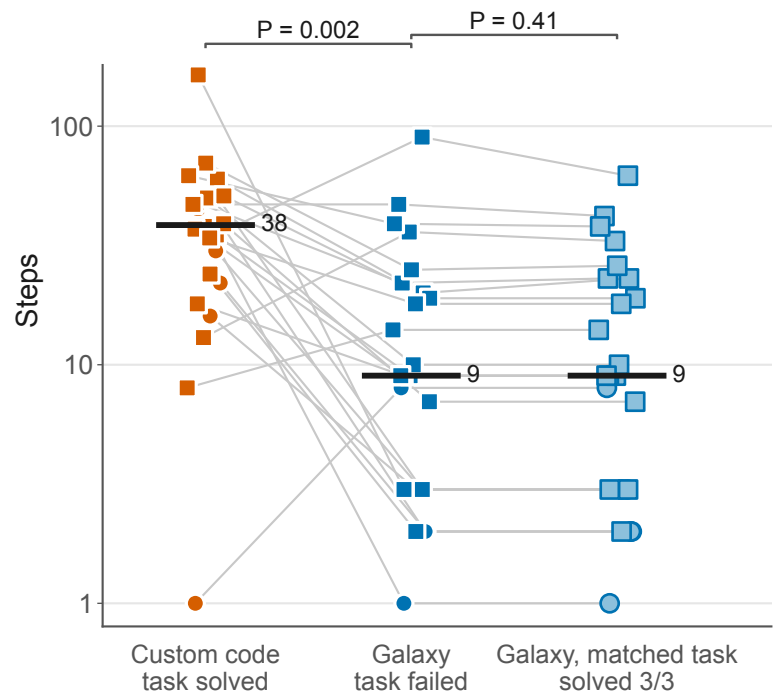
Path similarity

Same steps in the same order
(1 − normalised edit distance)



Steps per run

Median over the three replicates
(log scale)



Steps: Galaxy jobs (uploads, fetches and converters excluded) or shell commands (downloads, installs and file inspection excluded). Custom scripts are labelled by language and imported libraries, Galaxy user-defined tools by container. The matched controls make the Galaxy comparison fair in length; custom-code runs take more, finer steps, so their similarity is on a different scale.